

ch 05. Digital Logic.

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1. solution:

① Logic function: $F = \overline{AB} + AB$

② truth table

A	B	F
0	0	1
0	1	0
1	0	0
1	1	1

③ function: The circuit implement XOR gate. , or, we can say it can test whether A and B are equal.

2. solution:

$$11): F_1 = \overline{A}\overline{B}C + A\overline{B}C + BC = \overline{A}\overline{B}C + A\overline{B}C + A\overline{B}C + \overline{A}BC = m_1 + m_4 + m_7 + m_3$$

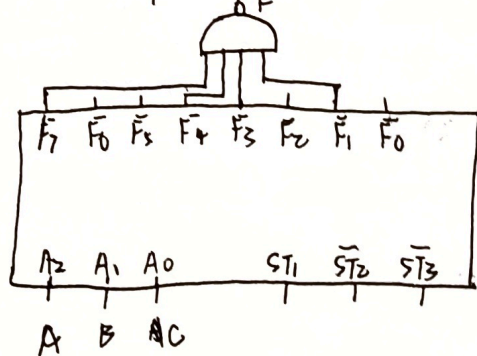
$$= \sum m(1, 3, 4, 7)$$

$$= \overline{m_1 + m_3 + m_4 + m_7}$$

$$= \overline{m_1 \cdot m_3 \cdot m_4 \cdot m_7}$$

$$= \overline{\overline{F_1} \cdot \overline{F_3} \cdot \overline{F_4} \cdot \overline{F_7}}$$

use 74LS138 to implement



$$12) F_2 = \bar{B}\bar{C} + A\bar{B}\bar{C}$$

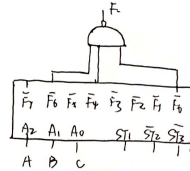
$$= A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C}$$

$$= m_4 + m_0 + m_6$$

$$= \overline{m_0 + m_4 + m_6}$$

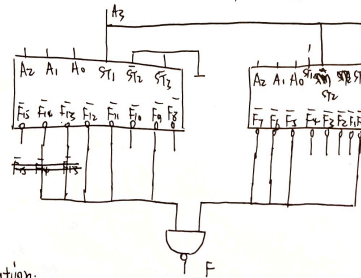
$$= \overline{m_0 \cdot m_4 \cdot m_6}$$

$$= \bar{F}_0 \cdot \bar{F}_4 \cdot \bar{F}_6$$



$$3. \text{ solution: } F = \sum m(1, 5, 6, 7, 9, 11, 12, 13, 14)$$

$$= \bar{F}_1 \cdot \bar{F}_5 \cdot \bar{F}_6 \cdot \bar{F}_7 \cdot \bar{F}_9 \cdot \bar{F}_{11} \cdot \bar{F}_{12} \cdot \bar{F}_{13} \cdot \bar{F}_{14}$$



4. solution:

use 4-bit full adder to implement Convert excess-3 to 8421 BCD code.

The relationship: Assuming that: $D_3 D_2 D_1 D_0$ is 8421 BCD code
 $C_3 C_2 C_1 C_0$ is excess-3 code

$D_3 D_2 D_1 D_0$				$C_3 C_2 C_1 C_0$			
C_3	C_2	C_1	C_0	D_3	D_2	D_1	D_0
0	0	0	0	0	0	1	0
0	0	0	1	0	1	1	0
0	0	1	0	0	1	1	1
0	0	1	1	0	1	0	1
0	1	0	0	0	1	0	0
0	1	0	1	1	1	0	0
0	1	1	0	1	1	0	1
0	1	1	1	1	1	1	1
1	0	0	0	1	0	1	0
1	0	0	1	1	0	1	1

For D_3 :

$C_3 C_2 \backslash C_1 C_0$	00	01	11	10
00	0	0	0	0
01	0	1	1	1
11	x	x	x	x
10	1	1	x	x

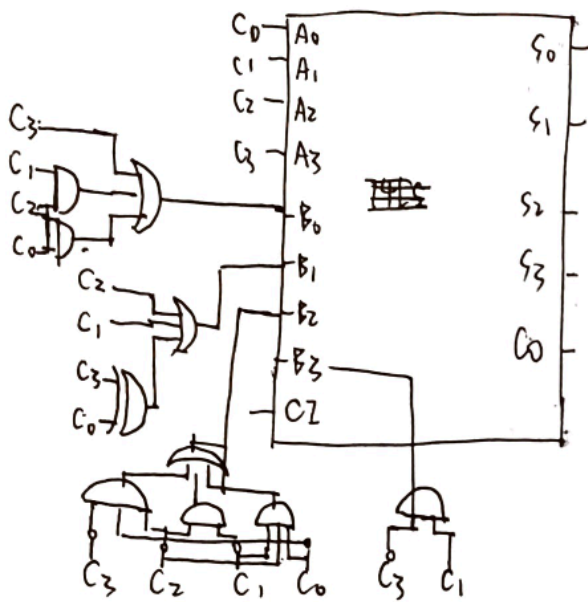
$$D_3 = C_3 + C_2 C_0 + C_2 C_1$$

Same: $D_2 = C_2 + C_1 + C_3 \bar{C}_0 + \bar{C}_3 C_0$

$$D_1 = \bar{C}_2 \bar{C}_1 + \bar{C}_3 \bar{C}_2 \bar{C}_0 + C_2 C_1 C_0$$

$$D_0 = \bar{C}_3 C_1$$

The Final Diagram



5. $M=0$. if input A, B, C all are 1, then $F=1$

$M=1$ if the majority of ABC are 1s. $F=1$.

other wise. $F=0$

(1) use 8-1 MUX 74LS 151 to implement

(2) use 3-8 M Decoder 74LS 138 to implement

Solution:

the truth table is

M	A	B	C	F
0	1	1	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0

the logic expression:

$$F = \bar{M}ABC + M\bar{A}BC + M\bar{A}\bar{B}C + M\bar{A}B\bar{C} + MABC$$

$$(1) F = \bar{A}_2\bar{A}_1\bar{A}_0D_0 + \bar{A}_2\bar{A}_1A_0D_1 + \bar{A}_2A_1\bar{A}_0D_2 + \bar{A}_2A_1A_0D_3 \\ + A_2\bar{A}_1\bar{A}_0D_4 + A_2\bar{A}_1A_0D_5 + A_2A_1\bar{A}_0D_6 + A_2A_1A_0D_7$$

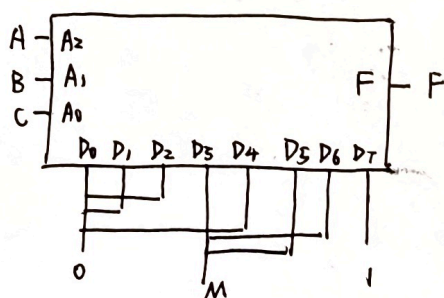
$$\text{Let: } A_2 = A, A_1 = B, A_0 = C$$

$$F = \bar{A}BC(M) + A\bar{B}C(M) + AB\bar{C}(M) + ABC(1)$$

$$\text{let: } D_3 = D_5 = D_6 = M, D_7 = 1$$

$$D_0 = D_1 = D_2 = D_4 = 0$$

Diagram



(2) Use the 3-8 Decoder.

$$F = \bar{A}BCM + A\bar{B}CM + AB\bar{C}M + ABC\bar{M} + ABCM$$

$$= \sum m(7, 11, 13, 14, 15)$$

$$= \overline{m_7 + m_{11} + m_{13} + m_{14} + m_{15}}$$

$$= \overline{\bar{F}_7 \cdot \bar{F}_{11} \cdot \bar{F}_{13} \cdot \bar{F}_{14} \cdot \bar{F}_{15}} = \overline{\bar{W}_7 \cdot \bar{W}_{11} \cdot \bar{W}_{13} \cdot \bar{W}_{14} \cdot \bar{W}_{15}}$$

Expand it to 4-16 decoder, and then get the diagram.

